Dr. Mauro Bernardini

Physicist and Data Scientist with a passion for data visualization

Nationality	German, Swiss, Italian, Spanish
Birthday	27.1.95
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PROFILE

I am an **Astrophysicist** with experience in **High-Performance Computing** (HPC) **Data Science** and numerical fluid simulations, based in Berlin Germany. My expertise lies in designing **visualization** strategies, **Deep Learning** applications and **AI integration** for **big data** pipelines.

EXPERIENCE

Dec 2023 - Dec 2025

Postdoctoral Researcher @ University of Zurich

- **Advanced** knowledge of **HPC facilities** such as CSCS (Switzerland) and LUMI (Finland)
- **Planning** and leading projects, writing proposals for project grants
- **Collaborating** in international groups and research projects
- **Supervising** of master students for semester projects, **teaching** assistant for physics and numerical courses
- **Presenting** at global conferences and workshops, **organizing** seminars and discussion panels

Nov 2023

Academic-Industry Exchange @ Meteomatics

- Developed a **temporal** image-based **super resolution** method for weather data
- Expanded professional network interfacing academia and industry partners

Oct 2023

Academic-Industry Exchange @ Ernst & Young

- Developed **Machine Learning based decision rules** to identify fraudulent **financial transactions**
- Gained experience on designing simple but compelling story-telling from complex datasets for industry partners

Sep 2019 - Sep 2023

Ph.D. candidate in Data Science @ University of Zurich

- Developed **Deep Learning** emulators for physics simulations
- Designed analysis pipelines for **big data** volumes → Main developer of visualization package for particle cloud data (link)
- Developing compelling **scientific story-telling** for diverse audiences

SKILLS

Soft
Critical thinking, scientific writing, public speaking, teaching, inclusive team/ project management

Languages
German (native), English (proficient), Italian (fluent), French, Spanish (basic)

Programming
Python, C, C++, SQL, Bash, HTML, CSS

Frameworks
Git, SSH, Conda, VS-Code, TensorFlow, PyTorch, Plotly, LaTEX

EDUCATION

Oct 2019 - Oct 2023	**Ph.D. in Data Science and Astrophysics** @ University of Zurich
Sep 2017 - Sep 2019	**M.Sc. in Astrophysics** @ University of Zurich
Sep 2013 - Sep 2017	**B.Sc. in Physics** @ University of Zurich

HOBBIES

Calisthenics, Windsurfing, Motorcycling, Art and Design, Fantasy and Science-Fiction literature